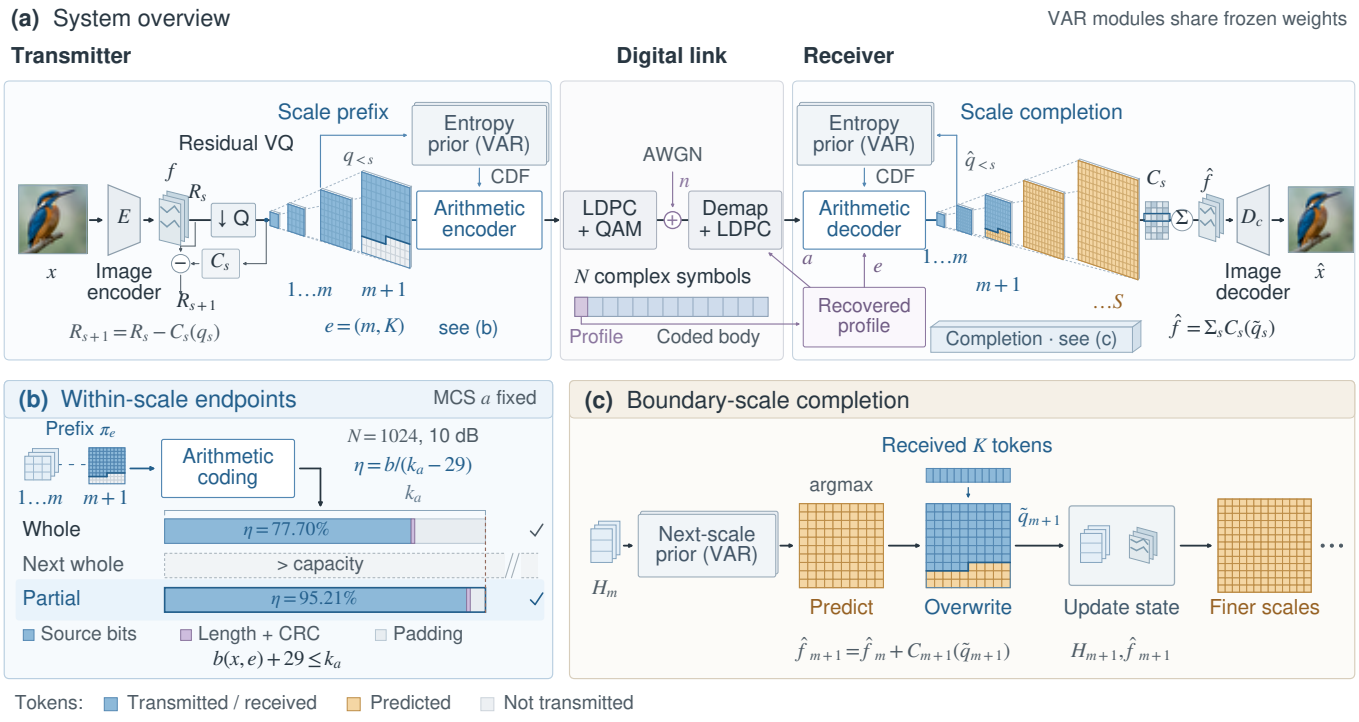

(a) System overview
VAR modules share frozen weights
Transmitter
Digital link
Receiver
Scale prefix
Residual VQ
Entropy prior (VAR)
CDF
Arithmetic encoder
Image encoder
x
E
f
R_s
↓ Q
C_s
R_{s+1}
$q_{<s}$
$1\ldots m$
$m+1$
$e=(m,K)$
see (b)
$R_{s+1}=R_s-C_s(q_s)$
AWGN
n
LDPC + QAM
Demap + LDPC
N complex symbols
Profile
Coded body
Entropy prior (VAR)
CDF
Arithmetic decoder
Scale completion
$\hat{q}_{<s}$
a
e
Recovered profile
$1\ldots m$
$m+1$
$\ldots S$
Completion · see (c)
C_s
$\hat{f}$
D_c
Image decoder
$\hat{x}$
$\hat{f}=\Sigma_s C_s(\tilde{q}_s)$
(b) Within-scale endpoints
MCS a fixed
Prefix π_e
$1\ldots m$
$m+1$
Arithmetic coding
$N=1024$, 10 dB
$\eta=b/(k_a-29)$
k_a
Whole
$\eta=77.70\%$
Next whole
> capacity
Partial
$\eta=95.21\%$
Source bits
Length + CRC
Padding
$b(x,e)+29\le k_a$
(c) Boundary-scale completion
Received K tokens
H_m
Next-scale prior (VAR)
argmax
Predict
Overwrite
$\tilde{q}_{m+1}$
Update state
Finer scales
$\hat{f}_{m+1}=\hat{f}_m+C_{m+1}(\tilde{q}_{m+1})$
$H_{m+1},\hat{f}_{m+1}$
Tokens:
Transmitted / received
Predicted
Not transmitted